



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

BACHELOR OF COMPUTER SCIENCE (DATA ENGINEERING) WITH HONS.

FACULTY OF COMPUTING

UNIVERSITI TEKNOLOGI MALAYSIA

SECP3213 - BUSINESS INTELLIGENCE

SECTION 02

PROJECT (PHASE 3)

GROUP : SKINAQUA

PREPARED BY :

NAME	MATRIC NO
YASMIN BATRISYIA BINTI ZAHIRUDDIN	A23CS0201
AIN NURNABILA BINTI MOHD AZHAR	A23CS0207
SYARIFAH DANIA BINTI SYED ABU BAKAR	A23CS0183
NURUL ADRIANA BINTI KAMAL JEFRI	A23CS0258

PREPARED FOR:

DR. NOORFA HASZLINA BINTI MUSTAFFA

TABLE OF CONTENT

1.0 DASHBOARD 1	3
2.0 DASHBOARD 2	4
3.0 KEY INSIGHTS	5
4.0 BUSINESS RECOMMENDATIONS	6

1.0 DASHBOARD 1

Figure 1.0 shows a dashboard of Sales KPI and Trend overview. The dashboard retrieves from the dataset that has been generated in phase 2 and connects it through data modelling. By having this dashboard, the user knows the current total revenue, total units sold, revenue per unit, product count, comparison between the total revenue by month and product tier, and trends of the total revenue by month. This dashboard could be viewed by selecting the region, category, and product tier filters.

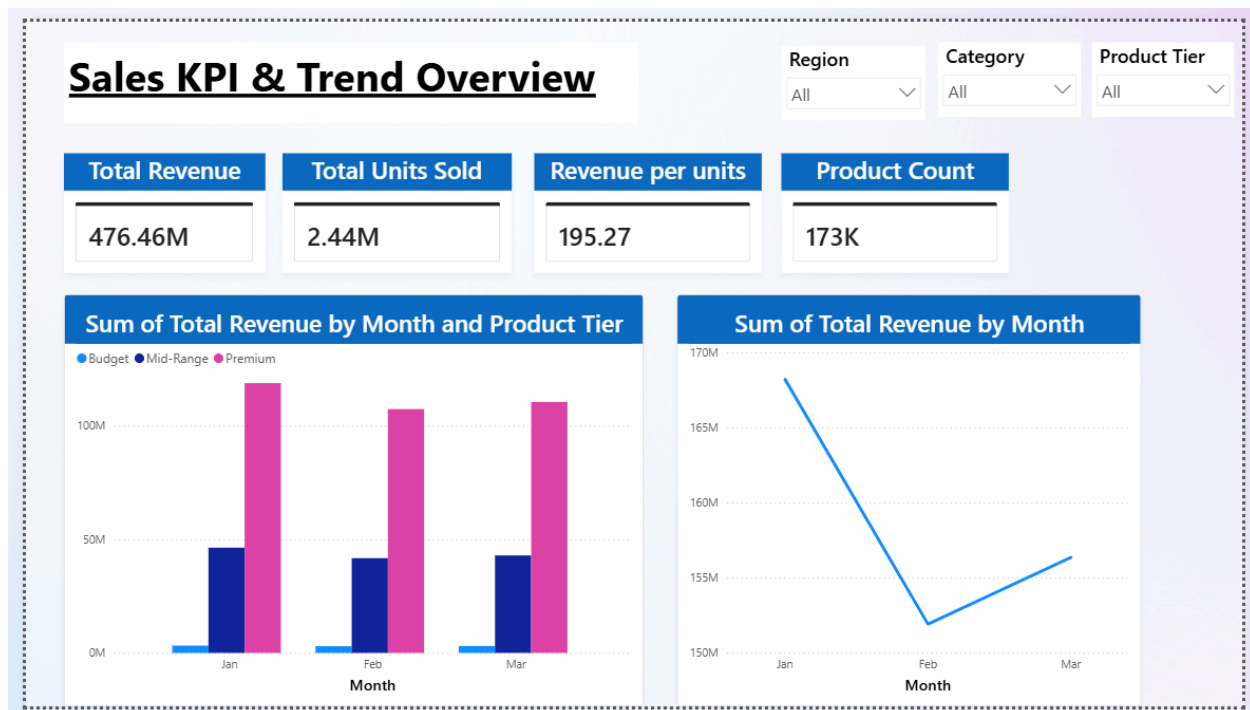


Figure 1.0 Dashboard 1

2.0 DASHBOARD 2

Dashboard 2 compares performance across product categories, regions, seasons, and discount strategy as shown in Figure 2.

1. Revenue vs Demand by Category

A column chart showing Total Revenue per category, with a line overlay showing the Average Demand Index per category.

2. Discount Rate vs Demand by Product Tier

A scatter chart plotting average discount rate against average demand index for each Product Tier (Budget, Mid-Range, Premium), with bubble size representing total revenue.

3. Region × Category Revenue Heatmap

A matrix showing total revenue for each region-category combination, with color shading from light to dark indicating relative performance.

4. Revenue Share by Season

A donut chart showing the percentage split of total revenue between Winter and Spring.

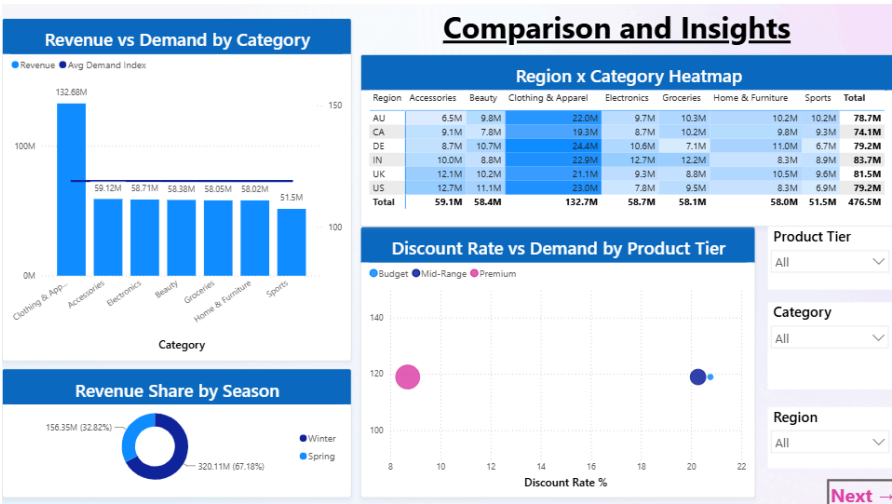


Figure 2.0 Dashboard 2

3.0 KEY INSIGHTS

1. **Premium revenue accounts for the vast bulk of revenues each month:** In January-March, Premium products remained at about \$100M/month, almost double that of Mid-Range (\$50M-\$60M), and well outnumbering the revenue of Budget, which is near-invisible. This is a month-over-month, non-seasonal tier ranking.
2. **Revenue fell off significantly in February before recovering in March:** Total monthly revenues dropped by about 10.6% from a total of \$170M in January to \$152M in February, but then rose back to \$157M in March. The dip is clearly seen on both the Mid-Range and Premium tier bars, indicating that it's not just a problem at a single tier but across both.
3. **Winter contributes 67.18% of the overall revenues:** The estimate for Winter was \$320.11M with \$156.35M for Spring (32.82%) directly confirmed. This is because Winter has two or three months in the data window (Jan & Feb) compared to one month for Spring (March) but has also suffered from a dip in one month (February) as mentioned above so the lead is not from a strong performance on the whole for Winter.
4. **There is no region which is not dominated by Clothing & Apparel:** Clothing & Apparel was the highest grossing category, with total sales of over \$132.68M, more than doubling the next largest category, Accessories (\$59.12M). The Region x Category Heatmap confirms this is not the norm: Clothing & Apparel is the highest-revenue category in all six regions, with \$19.3M generated in Canada and up to \$24.4M generated in Germany. On the Revenue vs Demand by Category chart, meanwhile, the Average Demand Index line remains relatively unchanged for all categories, indicating that Clothing & Apparel's revenue advantage is not on par with the same boost in demand.
5. **The higher the discount on Mid-Range, the higher the demand is not necessarily:** Premium products are discounted at $\approx 14\%$ and their demand index (≈ 120) is very close to Mid-Range products which are discounted at a greater margin of $\approx 20\%$. The difference in discount depth of about 6% does not result in a significant increase in the demand response in Mid-Range, indicating that

discount depth is not a major factor of the demand for these segments. Note: Actual percentages of discount will be taken off the chart and are interpreted by using the Power BI tippy tip before they are applied to the report.

6. **Only Germany and India have performed well in the region, while Canada has fallen short in all areas:** The highest regional total is from India (IN) and the UK (\$81.5M) is not far behind. Germany (DE) has the highest single region-category combination in the heatmap (\$24.4M in Clothing & Apparel) and is tied for second-highest regional total with the US (\$79.2M). Canada (CA) has the lowest regional total (\$74.1M), and lags in almost every category band, suggesting that the market is soft overall, and not just in one particular category.

4.0 BUSINESS RECOMMENDATIONS

1. **Re-evaluate discount strategy on Budget & Mid-Range tiers**

They're discounted nearly 2.5x more than Premium (~20% vs. ~8%) without producing higher demand, that margin could likely be redirected toward more effective growth levers.

2. **Investigate and replicate what's working in Clothing & Apparel**

It generates more than double the revenue of any other category despite comparable demand levels, suggesting stronger pricing, merchandising, or marketing execution that could be applied to underperforming categories.

3. **Dig into the January–February revenue dip**

Understanding what drove the mid-quarter slump (seasonality, inventory gaps, marketing timing) would help prevent or shorten similar dips in future quarters.

4. **Prioritize inventory and marketing spend ahead of Winter**

It drives 67% of total revenue, so stockouts or under-marketing during this window carry outsized risk to overall performance.

5. **Study high-performing region-category combinations to close regional gaps**

IN leads overall and dominates in Clothing & Apparel; CA lags behind, understanding what differs (pricing, assortment, local marketing) could help lift weaker regions.